THE COST OF CAPITAL

Learning Outcome

- calculate and interpret the weighted-average cost of capital for a company

As discussed in prior lessons, issuers make capital investments that are expected to have a return on investment greater than the required rate of return. An *issuer's* required rate of return on its capital investments is derived from its *investors'* required rates of return, adjusted for specific risks in the project. If the issuer has exhausted its positive-NPV project opportunities, it should return capital to investors so they can invest elsewhere and earn their required rate of return. Not doing so would destroy value for investors.

For an issuer, its required rate of return is also known as its **cost of capital**. Since issuers use both debt and equity, the cost of capital is composed of the **cost of debt** and the **cost of equity**. As described in prior lessons, debt is less risky than equity for investors because it is a priority, fixed claim on a firm's cash flows while equity is a residual claim of indeterminate length. Additionally, debt is sometimes secured with an underlying asset that would be transferred to the debtholder in the event of default, further reducing risk for the debt investor. Therefore, debt investors have lower required rates of return than equity investors because they take less risk; equivalently, an issuer's cost of debt is lower than its cost of equity financing.

An issuer's **weighted-average cost of capital (WACC)** blends its costs of debt and equity to obtain a single cost of capital. The WACC, after adjusting for any project-specific risks, is what issuers use as *r* in NPV analysis and as the hurdle rate for IRR analysis.

The calculation of WACC is shown in Equation 1:

$$\text{WACC} = (\text{Cost of debt} \times \text{Weighting of debt}) + (\text{Cost of equity} \times \text{Weighting of equity}) \quad (1)$$

The estimation of each component of Equation 1 will be covered in much greater detail in lessons on fixed income and equity later in the curriculum. For now, note the following observations.

- The cost of debt for an issuer is debt investors' required rate of return on debt financing. The interest rate on existing unsecured loans and bonds is typically a good starting point. However, if we want a forward-looking measure, meaning the interest rate on *new* debt raised to finance a new project, we might instead look to what interest rates similarly situated companies have recently borrowed at. Additionally, in jurisdictions where interest expense is tax deductible, we reduce the nominal cost of debt by multiplying by (1 – Tax rate) to obtain an after-tax amount.
- The weightings of debt and equity are either their market value proportions or target weights provided by management, which are typically on a book value basis. Market value weights are more commonly used because book values reflect historical prices of debt and equity, while investors' opportunity costs are based on current market prices of debt and equity.
- The cost of equity for an issuer is equity investors' required rate of return. Unlike the cost of debt, we do not have a historical interest rate to observe as a starting point. We do know that the cost of equity is higher than the cost of debt because of equity's riskiness and because distributions to shareholders are not tax deductible.

One simplistic approach is to simply observe the historical returns on equities in general—because perhaps equity investors are expecting a similar rate of return going forward—and adjust that rate for the company-specific considerations we will discuss in the next lesson, such as how stable a company's cash flows are.

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of equity we'll use for these lessons. Expected and required rates of return for equity investors are rich topics that will be discussed in more detail later in the curriculum.

- If an issuer has additional types of financing with different risk and return characteristics from those of debt and equity—for example, preferred stock and non-controlling interests—the weighted-average after-tax cost of that financing needs to be added to Equation 1 as well.

Knowledge Check: Computing WACC

1. Assume that ABC Corporation is financed with 40% debt and 60% equity. Also assume that interest expense is tax deductible. ABC Corporation wishes to maintain these proportions as it raises new funds.

   ABC's debt investors' required rate of return, measured before taxes, is 4%, and its equity investors' required rate of return is 10%. If the company's marginal tax rate is 23%, what is ABC's weighted-average cost of capital?

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   Solution:

   ABC's weighted-average cost of capital is 7.23%:

   WACC = [(1 – Tax rate) × Pre-tax cost of debt × Weighting of debt] + (Cost of equity × Weighting of equity).

   WACC = [0.04 × (1 – 0.23) × 0.40] + (0.10 × 0.60).

   WACC = 7.23%.

Discussion

If ABC's management maintains the company's current mix of financing (40% debt, 60% equity), what would happen to ABC's weighted-average cost of capital if its cost of debt decreases by 1%? If its cost of equity increases by 1%? What about if the company's tax rate changes? Write your responses on this lesson's discussion board on the LES.

As demonstrated in the prior module, capital investments increase firm value if their returns exceed required rates of return. The prior module focused on the return on investment as the main driver of value, but it is also clear that a lower required rate of return would increase NPV, too, or at least provide a lower threshold for value-creative investments. This forms the basis for one of the most common objectives for managers in **capital structure** decisions, the mix of debt and equity financing: Choose whichever one leads to the lowest WACC. However, the cost of debt and equity are not chosen by management but, rather, are determined in financial markets by investors, so it is a dynamic and non-trivial endeavor. The second objective of management in choosing a capital structure is, where possible and economical, matching the liquidity or time horizon with that of its capital investments.

Question Set

expense is tax deductible. Debt investors require a before-tax return of 6%, and equity investors' required return is 12%. If the marginal corporate tax rate rises from 20% to 25%, the change in the WACC is closest to:

A. –0.09%.
B. 0.09%.
C. 0.30%.

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Solution:

A is correct. XYZ's WACC at a tax rate of 20% is calculated as follows:

WACC = (Weighting of debt × Cost of debt) + (Weighting of equity × Cost of equity)

= (0.3)(6%)(1 – 0.2) + (0.70)(12%) = 9.84%.

XYZ's WACC at a tax rate of 25% is calculated as follows:

WACC = (0.3)(6%)(1 – 0.25) +(0.70)(12%) = 9.75%.

Thus, WACC declines by 0.09% as the after after-tax cost of debt declines from 4.8% to 4.5%.

2. True or False: Since the cost of debt and equity are controlled by the firm, managers can lower the cost of capital for the firm by reduce reducing the cost of debt and equity used in their capital allocation process.

A. True
B. False

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Solution:

B is correct. This statement is false because the costs of debt and equity are not selected by management but are determined in financial markets by investors, though they are influenced by management's actions.

Questions 3 and 4 relate to the following information:

Company X has the following on the right-hand side of its balance sheet:

Bonds $400,000

Common stock (40,000 shares) $600,000

Total liabilities and equity $1,000,000

The price of Company X stock is currently $20 per share. The required return before tax on debt is 5%, and the required return on equity is 10%.

3. In calculating WACC for Company X, the weight for debt using book values for the balance sheet components is likely to be _____ than the weight for debt using market values for the balance sheet components.

A. lower
B. the same
C. higher

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Solution:

C is correct.

The book value weights are computed as follows:

W_d = $400,000/$1,000,000 = 0.40.

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The market value of equity is equal to $25 × 40,000 shares = $1,000,000.

The market value weights are computed as follows:

W_d = $400,000/$1,200,000 = 0.33.

W_e = $800,000/$1,200,000 = 0.67.

Thus, the book value weight for debt (0.40) is higher than the market value weight (0.33).

4. If the marginal corporate tax rate is 20% and the stock price increases to $25, the WACC for Company X is closest to:

A. 7.60%.
B. 8.26%.
C. 8.55%.

Hide Solution

Solution:

B is correct.

The book value weights are computed as follows:

W_d = $400,000/$1,000,000 = 0.40.

W_e = $600,000/$1,000,000 = 0.60.

The market value of equity is equal to $25 × 40,000 shares = $1,000,000.

The market value weights are computed as follows:

W_d = $400,000/$1,400,000 = 0.29.

W_e = $1,000,000/$1,400,000 = 0.71.

The appropriate weights for the WACC calculation are the market weights

Company X's WACC, using the market weights, is calculated as follows:

WACC = (weighting of debt * cost of debt) + (weighting of equity * cost of equity)

= (0.29) (5%)(1-0.2) +(0.71)(10%) = 8.26%

5. Fill in the blanks using the following two terms:

higher

lower

Shareholder value is increased by a _____ return on investment and a capital structure that results in a _____ WACC.

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Solution:

Shareholder value is increased by a *higher* return on investment and a capital structure that results in a *lower* WACC.

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MD

WACC with debt at 3% = 6.924%

WACC with equity at 11% = 7.832%

WACC has inverse relationship with changing tax rates: WACC up if tax rates down since less interest tax deductible, and vice vers

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HC

WACC = (Cost of debt × Weighting of debt) + (Cost of equity × Weighting of equity)

(0.4) (3) (1-0.23) + (0.6) (7)

= 7.9240

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PC

If cost of debt decreases by 1%, WACC goes down by 0.31%. If cost of equity goes up by 1%, WACC goes up by 0.6%. If tax rate increases, WACC goes down and if it decreases then WACC goes up.

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KS

Revised WACC will be = (0.60*0.10)+((0.30*0.77)*0.40) = 6.92%.

So, WACC fall by (7.23-6.92%) = 0.31%

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AS

ABC case: with 40% debt, 60% equity, a 4% before-tax cost of debt, a 10% cost of equity, and a 23% tax rate, the WACC is about 7.23%.

If the cost of debt drops by 1 percentage point (to 3% before tax), WACC falls to ≈6.92% (−0.31 pp).

If the cost of equity rises by 1 percentage point (to 11%), WACC increases to ≈7.83% (+0.60 pp).

If the tax rate moves from 23% to 25%, the stronger interest tax shield nudges WACC down to ≈7.20% (−0.03 pp).

Takeaways: WACC is more sensitive to the cost of equity than to equal-size moves in the cost of debt; a higher tax rate slightly lowers WACC; cheaper or greater use of debt typically reduces WACC but increases financial risk.

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AP

Previously, I've calculated, the increase of debt by 1% not the decrease (as it was asked).

In case, the debt decreases by 1% from 4% to 3%..

The WACC will decrease, as follows:

0,03*(1-0,23)*0,4+0,6*(0,1)=6,92%

Under the assumption that weights for debt and equity remain the same

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AP

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AP

If cost of debt rises by 1%, WACC rises from 7,23% to 7,54%. Under the assumption that weights for debt and equity remain the same

0,05*(1-0,23)*0,4+0,6*0,1=7,54%

If cost of equity rises for 1%, we have to take into account that the weight of equity is 60% (higher than the weight of debt), therefore WACC not

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Finally if corporate tax changes, will depend if it increases or decreases.

If corporate tax rate increases, the WACC will decrease, because (1-tax rate) is lower. However, if corporate tax decreases WACC will increase because (1-tax) is higher

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XT

Under the scenario where required rate of return for debt = 4%; equity investors' required rate of return = 10% and with company's marginal tax rate = 23%, we obtain the weighted-average cost of capital = 7.23%.

If the cost of debt decreases by 1%, we will expect a decrease in the cost of capital.

WACC = 40% x 77% x 3% + 60% x 10% = 6.92%

If the cost of equity increases by 1%, we will expect an increase in the cost of capital.

WACC = 40% x 77% x 4% + 60% x 11% = 7.83%

If the company's tax rate changes, it will affect the after-tax cost of debt. In the case where the company's tax rate increases, the after-tax cost of debt will decrease and thus the cost of capital will decrease. Conversely, when the company's tax rate decreases, the cost of capital will increase.

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